DENKLEMLER

ÖRNEK $3x-5=0$ denkleminin çözümünü bulunuz.

$\Rightarrow 3x-5=0$ (5 karşıya +5 olarak geçer)

$3x=5$

$\frac{3x}{3}=\frac{5}{3} \Rightarrow x=\frac{5}{3}$ //

ÖRNEK $3x+12+x-8=10-3x+8$ denklemini çözünüz.

$\Rightarrow 3x+12+x-8=10-3x+8 \rightarrow 4x+4=18-3x$

$\rightarrow 7x=14$

$\rightarrow x=\frac{14}{7}=2$

$\rightarrow Ç.K=\{2\}$

ÖRNEK $\frac{1}{2x}+\frac{3}{x-5}=0$ denklemini çözünüz.

$\Rightarrow \frac{1}{2x}+\frac{3}{x-5}=0 \Rightarrow \frac{x-5+6}{2x(x-5)} \Rightarrow 0$

$(x-5)\ (2x)$

$\Rightarrow \frac{7x-5}{2x(x-5)}=0$

$x=0 \quad x=5$

$\Rightarrow 7x-5=0 \rightarrow x=\frac{5}{7}$ $\quad Ç.K: \frac{5}{7}$

ÖRNEK $\left.\begin{matrix} 2x-y=-1 \\ x-2y=4 \end{matrix}\right\}$ denklem sisteminin çözümü nedir?

$2x-y=-1$

$-2 \,/\, x-2y=4$

$2x-y=-1$

$+\ -2x+4y=-8$

$3y=-9$

$y=-3$

$2x-y=-1 \Rightarrow 2x-(-3)=-1$

$\Rightarrow 2x+3=-1$

$\Rightarrow 2x=-4$

$\Rightarrow x=-2$ $\quad (x,y)=(-2,-3)$

ÖRNEK $\left.\begin{array}{l} x+6y=162 \\ \frac{y}{x}=\frac{1}{3} \end{array}\right\}$ denklem sisteminin çözüm kümesi nedir?

$x+6y=162 \Rightarrow 3y+6y=162$

$\Rightarrow 9y=162$

$\Rightarrow y=18$ //

$\Rightarrow \frac{y}{x}=\frac{1}{3} \Rightarrow x \cdot \frac{3}{y}$

$\Rightarrow \frac{18}{x}=\frac{1}{3} \Rightarrow x=54$ //

$\mathcal{Ç}=\{(54,18)\}$

ÖRNEK $\left.\begin{array}{l} a+\frac{1}{b}=4 \\ b+\frac{1}{a}=3 \end{array}\right\}$ $\Rightarrow \frac{a}{b}$ ifadesi neye eşittir?

$a+\frac{1}{b}=4 \Rightarrow \frac{ab+1}{b}=4 \Rightarrow ab+1=4b$

$b+\frac{1}{a}=3 \Rightarrow \frac{ab+1}{a}=3 \Rightarrow ab+1=3a$

$\Rightarrow 4b=3a \Rightarrow \frac{a}{b}=\frac{4}{3}$ //

ÖRNEK $\left.\begin{array}{l} 5x+2y=1 \\ 2x+y=1 \\ ax+3y=5 \end{array}\right\}$ denklem sisteminin a) çözümü nedir?

b) a'nın değeri kaçtır?

ⓐ $5x+2y=1$

$2x+y=1 \ /\cdot -2$

$\underline{\pm \ -4x-2y=-2}$

$x=-1$

$5x+2y=1 \Rightarrow 5.(-1)+2y=1$

$\Rightarrow 2y=6$

$\Rightarrow y=3$

$(x,y)=(-1,3)$

ⓑ $ax+3y=5 \Rightarrow a.(-1)+3.3=5$

$\Rightarrow -a+9=5$

$\Rightarrow a=4$ //